Pragmatics of the Diagram

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Diagrams are an integral part of modern life. For the cover design of the architectural magazine Oase's diagram special issue, the Dutch graphic designer Karel Martens used a photograph of an ad-hoc 'map' that his son drew with the objects on the table. Image 1. It reveals a form of diagrammatic communication, fully embedded in our daily life – although a diagram is not always recognized as such. Most often, diagrams are charts, graphs, drawings or figures that explain the facts of particular cases, and a diagram in this sense is easily recognizable for its generic form. As one of the oldest forms of graphic communication, the diagram has been an important element in the development of modern graphic design. Jan Tschichold's typography in its early years often adopted diagrammatic devices to articulate the meaning and the structure of a text, Image 2. while in Otto and Marie Neurath's Isotype diagram was elevated to the level of international – thus universal – picture language. Image 3; see Otto Neurath, International Picture Language, originally published in 1936 [the facsimile edition is available from Reading University Department of Typography & Graphic Communication]. One may go even further to claim that, without the diagram and its capacity to precisely represent numbers and facts, the entire history and practice of 'information design' would have been impossible. In this respect, the diagram naturally falls in a graphic designer's interest.

The diagram is also a means to understand the world, which would otherwise remain too complex to comprehend. In the area of scientific research, it has become an essential tool to describe complex phenomena, and this visual description and articulation has often yielded unexpected revelation. Image 4. Furthermore, some philosophers recently began to view the diagram as not only a tool for explanation, but also an active entity that shapes the world according to its own logic. Thus when Michel Foucault called the panopticon as the diagram of power, Image 5. he did not use the term just metaphorically; it is a fundamental, if abstract, matrix that actively manifests itself in political and juridical organizations and practices – an idea that found a different expression in Gilles Deleuze's notion of the 'abstract machine'. Thus, the architectural theorist Sanford Kwinter states: [Diagrams] are abstract because conceptually and ontologically distinct from material reality yet they are fully functioning machines nonetheless, that is, they are agencies of assemblage, organization and deployment. Reality, to speak a bit reductively, is comprised both of matter and the organization of that raw matter into deployable objects or complexes. The argument, stated simply is as follows: to every organized entity there corresponds a micro-regime of forces that endows it with its general shape and program. Every object is a composition of forces, and the compositional event is the work or expression of an abstract machine. Sanford Kwinter, 'The Hammer and the Song', Oase, no.48, p.34.

The focus of this investigation lies somewhere between these two notions of the diagram. It will be firmly based on the functions of the diagrammatic in graphic design and visual arts, without overly trying to address its 'ontological status' in the philosophical sense; at the same time, however, it will not be committed to simply examining the 'true' meanings of the diagram as a graphic tool or exalting its 'proper' functions. It is quite possible, one may argue, to investigate other uses of diagrams as means to persuade, to allude, to empower, to tell a story, to render or provoke imagination, or simply to generate forms.

Diagram as a Rhetorical Device The diagram has a strong association with 'facts', largely accepted as a rational, neutral and objective way of depiction. Apparently indifferent to any partial interest of the content, its sheer visual form alone can overwhelm the viewer with the weight of truth. It is this authority and alleged legitimacy that renders the diagram so effective as a

means of persuasion. If any figure of rhetoric must struggle to be convincing by disguising its 'intention', the diagram does not appear offensive or manipulative: after all, it is a set of 'facts' – whether you like it or not.

No wonder, then, that the statistic diagram has been heavily used in advertising to render the products more convincing. Stefan Sagmeister's design for the 'Move Our Money' campaign is simply a recent, if slightly unconventional, contribution to this long-lasting tradition. Image 6. Sagmeister built a set of mobile air-balloon 'sculptures' in the shape of graphs, pie charts and Isotype figures, to illustrate the government's tax expenditure. The disproportion of the us military expenditure is revealing, but the sense of absurdity is reinforced by the graph's comically unstable physical presence.

But even in the seemingly neutral area of information design, diagrams can perform rhetorical functions, perhaps in a more insinuating way. In his essay 'The Rhetoric of the Neutrality', Robin Kinross analyses diverse examples from railroad timetables to equipment catalogues, Image 7. and how different design of the same function could encode different cultural references. For example, in 1968 Gui Bonsiepe illustrated his design method based on then newly-emerging information theory by re-designing an existing industrial equipment catalogue. Replacing photographs in the catalogue with clear-cut diagrams was part of his new approach. Suggestively, Bonsiepe concluded that his new design was not only more rational and ordered than the original, but more 'beautiful'. Quoted in Robin Kinross, 'The Rhetoric of the Neutrality', in: Victor Margolin (ed.), Design Discourse: History, Theory, Criticism, Chicago: University of Chicago Press, 1989.

Although not included in Kinross's list, Harry Beck's famous re-design of London Underground map in 1933 is another good example. Image 8. According to Adrian Forty's analysis in his book *Objects of Desire*, this new map promoted the efficiency of the by then newly established corporate body by adopting the appearance of rationalized forms, at the price of truer representation of the city's geographic reality. Adrian Forty, Object of Desire: Design and Society 1750–1980, London: Thames & Hudson, 1986. Although it might be argued that both examples are simply the results of following the proper logic of diagram and 'information theory', it cannot explain the fact that there can be a number of different designs that are, to some degree, all 'correct'. If Harry Beck's London Underground map is the correct answer to the problem, then how to justify the obviously different designs of, say, Italy or New York, other than by – questionably – suggesting their inferiority? There must be some room in this neutral area of design that is not determined by pure logic, and often occupied by rhetorical interests. The very fact that they are supposedly value-free diagrams makes it difficult to recognize this redundancy. Design is a social activity, not free from human interest and cultural values, so the rhetorical dimension may be intrinsic to its outcomes. Kinross asks, however, 'if nothing is free from rhetoric, what can be done to seem free of rhetoric?' Certainly, the diagram has been one answer.

Diagram as a Style Most contemporary diagrams share certain common visual characteristics: abstract or simplified shapes; structural clarity; clean, precisely drawn lines; pure and reductive composition. These formal features have been established as such mainly for functional reasons: to reduce visual 'noise', to eliminate redundant particulars from the elements, and to logically render the most essential relationship between the elements. One cannot ignore, however, the historical and ideological dimensions that affect their formation. Isotype, for example, had internationalist aspirations in its attempt to establish 'universal' visual language that could bypass cultural and historical barriers in the aftermath of the first World War in Europe. Image 9. Thus an Isotype 'man', without any personal or ethnic characteristics, was meant to represent the universal man – not a German, a Frenchman or an Englishman. In this sense, it was a visual equivalent of Esperanto. It is not difficult to imagine what the formal features of Isotype – and of diagrams for that matter –

could signify at that time: internationalism, equality, history- and culture-free modernity.

About eighty years later, the visual characteristics of the diagram frequently carry different connotations than the political ideals. As if to testify to the defeat of their original modernist aspirations that have gone wrong in the course of history, the diagram's pure and abstract forms are often appropriated to depict the cold and impersonal nature of modern world. NatWest Bank's TV commercial by TBWA GGT, for example, uses the diagrammatic visual language usually found in in-flight safety manuals, to encourages 'escape' from the routines of office work. Image 10. Some take this approach even further, using the diagrammatic to glamorize alienation, and render it fashionable. Kam Tang's illustrations for the fashion spreads of Arena magazine portray the impeccably dressed male model placed in hyper-technically drawn urban environments. Image 11. These images seem to suggest that the 'vertigo' often felt in these technological environments could be bearable or even desirable – in the designer-suits the magazine is promoting. Here, the diagrammatic becomes a style, a fetish.

Indeed, the fetish of the diagrammatic has become one of the major elements in many designers' portfolios, as exemplified in the spreads by the Zürich-based graphic design practice Norm. Image 12. The monochrome photograph of a group of Japanese scientists is suggestive: is it an ironic nostalgia for a bygone era when the rationalist, 'scientific' approach to design was at once necessary and fashionable, when science and technology were seen as an absolutely progressive and constructive force, and Japanese culture as its purest crystallization? The ironic attitude towards the diagrammatic resembles what has been called 'techno-orientalism', in which the Japanese are often represented as fashionably robot-like technocrats. For more information about 'techno-orientalism', see Toshia Ueno, 'Japanimation and Techno-Orientalism' in: Bruce Grenville (ed.), The Uncanny: Experiments in Cyborg Culture, Vancouver, Vancouver Art Gallery & Arsenal Pulp Press, 2001.

Proliferation of diagrammatic irony is a relatively recent phenomena, but not without precedents. One of the earlier examples, and indeed one of the most provocative, is the sleeve of Joy Division's album Unknown Pleasures (1979), designed by Peter Saville. Image 13. Its minimal design with a computer-generated diagram of eighty successive periods of the first pulsar observed quickly acquired a cult-status among fans. As a textbook example of graphic appropriation and in fact a formally unique graphic design piece of a time when the raw, expressive language of punk was becoming ubiquitous, the sleeve was also a perfect depiction of the band's music, which was often based on computer-generated, mechanical musical pulses.

Diagram and Power Diagrams signify power. They often use standardized conventions, which are legitimated by the authority. Their 'correctness' is always judged against those conventions and rules, which are to be learned through education. In this process, a hegemonic understanding of the world is shared, agreed and perpetuated. Pierre di Sciullo's greeting card for a humanitarian organization *Cause Première* encourages us to realize how our perception of the world depends on its 'official' diagram, by showing the upside-down map of France. Image 14. Indeed, cartography is an area where the diagram and politics are inseparably interwoven.

Simply drawing a line on the map can be a political action, with serious implications: it is not only about articulating geographic information, but also an expression of political will. A map is also indispensable as a tool for military and economical operations. It should come as no surprise that cartography has long been the most sophisticated and rigorous in the broad field of graphic arts, because making and having a useful map is oftentimes a matter of life and death.

Here, the diagram itself becomes power, by providing a kind of knowledge that can only be acquired from a privileged position, with control over its subject: thus, the struggles over building plan in many bank robbery movies, Image 15. the map of Iraqi territory and the plan of its Presidential Palace in news media, Image 16. the family tree of the Mafia in a triumphant police announcement. Image 17. The power a diagram promises, however, can be merely illusory, as shown in the movie Dr. Strangelove where the gigantic world-map in the war room which might have once provided the sense of control to the us decision makers now only accentuates their impotence. Image 18.

Diagram as a Narrative Device Diagram can be an effective means to tell a story, presenting events in time in a condensed graphic form. Charles Joseph Minard's classic diagram of Napoleon's campaign in Russia is a good example. Image 19. Two bands represents the paths of the army: the gray band shows the advance towards Moscow; the darker one that of its retreat. The width of the band indicates the size of the army. They are linked to a temperature scale at the bottom. One can instantly understand how the temperature affected their campaign, when and where the army suffered most, etc. Drawn in 1861, this diagram may well pale many recent experiments in the multiple narrative.

Diller and Scofidio's visual essay Case no. 00-17163 attempts a new 'genre' of diagram-laden storytelling. Image 20. It reconstructs a murder case investigation, combining texts, photographs of a place and various evidences, with diagrams of social relationship of the people involved, a timetable and graphic devices that link diverse elements. This rich and complex diagrammatic contribution to the book *Zone 6*: Incorporations encourages the reader to actively make a case from those visual and textual clues, in a similar way a detective story transforms its reader to an investigator.

Diagrams can be used as an effective editorial tool as well. The opening spreads of Rem Koolhaas and Bruce Mau's book S,M,L,XL run a series of statistical graphs that show the global distribution of studio projects, the changing size of the office and its income, the number of nights Koolhaas spent in hotel rooms and more. Image 21. The diagrams create an interesting contrast with the mundane pictures of Koolhaas's office in the background. This sequence of sheer facts speaks of the office's history and the breadth of their work, perhaps more eloquently than any elaborate prose.

Much like a satirical newspaper cartoon, a diagram can perform as a report or commentary, succinctly summarizing views on certain events or phenomena. Scott King's diagrams of two Sex Pistols' shows – at London in 1976 and San Francisco in 1978, respectively – capture an aspect of typical relationship between a rock band and its audience. Image 22. They not only demonstrate the growth of the band's popularity over a period of two years, but also suggest the inequality in relationship between the band and its audience. In these unbalanced fractional forms, with the dots above signifying the audience and those below the band members, the gap between the selected few on stage and many others on the Åoor becomes evident. It can also be seen as a commentary on the nature of popular culture, where only a few cultural icons mobilize the masses.

Diagram and Imagination Beyond relaying the facts, a diagram can also visualize imagination. Certainly, it is true that any technique of visual depiction is capable of representing the imaginary, but the diagram has unique qualities, different to, say, realistic imagery. A diagram will provide a coherent, rationalized description of a fictional being, while a realistic image tends to convey only its appearance: the diagram would leave the actual appearance to be completed in the viewer's mind. Thus, BBC Television used fictional diagrams when it was making a TV version of the legendary radio show, *The Hitch Hiker's*

Guide to the Galaxy. Image 23. Its simple yet vivid diagrams successfully translated the far-reaching imagination of the original by, paradoxically, leaving room for imagination to the viewers, instead of exhausting it by hyper-realistic imagery.

The Hitch Hiker's Guide to the Galaxy is science fiction, a genre that uses diagrams to render the fictional more likely. But in the history of real, 'hard' science too, one can witness many diagrams of assumed reality that have eventually turned out to be false. Medieval celestial maps in which the Sun is revolving around the Earth, or a T-O map that presents a wrong – at least to modern viewers – conception of continents are such examples. Image 24. They existed in between reality and imagination, negotiating between empirical facts and the yet unknown. Even today, some unprovable religious beliefs and speculations depend on diagrammatic representation for their social currency.

Even if not fictional by nature, some diagrams provoke imaginative responses from their viewers. Diagrams often use codes and conventions, and without precise knowledge of these, viewers might find them mysterious and enigmatic. Unlike completely abstract paintings, a diagram – or diagrammatic drawing – always seems to refer to something concrete, and when incapable of deciphering its meaning, one is urged to fill this gap with one's own imagination.

The diagram sent to the universe with the Pioneer spacecrafts, for example, was designed for a real and specific, if too ambitious, purpose: communicating with the alien intelligence. Image 25. The imaginative anticipation of 'Contact' and its precise yet enigmatic appearance have stimulated the imagination of many designers, including Todd St. John, who recently cited it as his major inspiration, 'Inspiration', Eye, no. 34. and Min Choi, who confessed his Monospace Font Test Pattern was a reverent tribute to the Pioneer diagram. Image 26.

Diagrams of similar effect can also be found in the world of fine arts. Marcel Duchamp's so-called Large Glass resembles the diagram of some kind of machinery, the function of which is mysterious. Image 27. Responses to this piece have been various, from this literally rendered animation, Image 28. to a more freely interpreted performance by Diller and Scofidio, which investigates every mechanical possibilities implicit in Duchamp's structure. Image 29. The fact that the artist Richard Hamilton had to produce two books – Green Book (1993) and White Book (1999) – about this piece as a way of understanding, only testifies to the profound enigma of the Large Glass. Image 30.

Diagram as a Form-Generating Machine Formal beauty or novelty is never the main concern of 'proper' diagrams. Certain diagrams, however, accidentally yield unexpected formal configurations, which might otherwise be difficult to envision. Some designers have exploited this possibility by systematically working with diagrams, instead of trying to create forms intuitively.

The American architect Laura Kurgan's design for the Museum of the Contemporary Arts in Barcelona is a good example. Image 31. By walking with a gps receiver, she generated a drawing that would be used as a plan for the building. The resulting built form might be called as a diagram of a walk in the networked space, realized as a concrete structure. The architect Peter Eisenman is also known to use diagrams to conceive and generate spatial forms. Contrary to the common notion of diagram as a clarifying device, Eisenman's diagrams often anticipate the form that defies easy and clear grasp. Eisenman himself has noted that a diagram can become 'both rational and mythical, a strange superposition of the two.' Peter Eisenman, 'Diagram: an Original Scene of Writing', Any, no.23.

In the field of graphic design, too, diagrams have been used as an engine, if not a mere excuse, to generate forms. The *American Center for Design's 32th Annual 100 Show* catalogue employs a series of diagrams that arrange the entries by different parameters – client, designers' location, size of work,

dominant color, and medium. Image 32. Despite the forced nature of some of these parameters, this presentation is certainly useful for understanding – although one may have some reservations on its readability. However, one can also suspect if the real motivation of these information-laden spreads was not in creating beautiful patterns and textures. This beauty also appeals to the intellect, as everything is seemingly motivated by meaning. There is a danger of using the diagram purely for formal reasons: it can be too beautiful to allow other responses to arise, or it can appear too meaningful without really providing any relevant reading. A diagram can be rhetorical, fashionable, powerful, telling, provocative, so it can also be simply beautiful. But one must go further than this, and ask: what makes it aesthetically so compelling? How exactly does it please our eyes and minds? In what context?

Diagram in Context These examples are not exhaustive, nor are the categories mutually exclusive. I used them simply to suggest how the diagram can perform different functions depending on their contexts. Struggling to enhance or simply beautify them, designers often overlook how the diagram actually works in the world; this essay is suggesting not overlooking them, but reading their function in their particular contexts. In linguistics, pragmatics deals with the function of language, rather than its grammatical constructions or literary value: to this view, 'correctness' or 'beauty' of a text becomes only relevant – or irrelevant – depending on its specific context. In a sense, this text has advocated similar attitude to understanding the diagram: sensitive to its various functions in contexts, attentive to their unexpected effects, a pragmatic understanding of the diagram.

Bibliography